



**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 3

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Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats researchers and entrepreneurs in the field of computers science and engineering.

Mission

- The impact quality education in computer science and engineering by means of learning techniques and value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting edge technologies, through industry interactions.
- To build leadership, skills, ethical values and team work among the students.
- To strengthen the collaboration of departments and industry internships, mentorship and professional body activities.

1. Define Anti virus program. What are the roles of a security team.

- * Anti virus software is a program or set of program that are designed to prevent, search for detect, and remove software viruses, and other malicious software like worms, trojans, adware and more.
- * These tools are critical for users to have installed and up to date because a computer without anti-virus software installed will be infected within minute of connecting to the internet.
- * There are several different companies that build and offer anti-virus software and what offers can vary but all perform some basic functions.
- * Scan specific files or directories for any malware known malicious patterns.
- * Allow to schedule scans to automatically run for you.
- * Allows to initiate a scan of specific file on the computer or of a cd, or flash drive at any time.
- * Shows the health of the computer.
- * Remove any malicious code detected will be notified of an infection and asked to clean the file, other programs will automatically do this behind the scans.

Viruses and worms

- * A new threat has arisen in the past few years to cause concern among data processing and relative worm.

- The entities range from harmless to destructive. A virus is a program that can affect others by modifying them: the modified program included a copy of the virus program.

Nature of viruses

- A computer virus carries in its instruction code the capability for making copies of itself.
- A virus can do anything that other programs do, the only difference is that it attaches itself to another program.
- If after a virus program is executed it can perform any function, such as erasing files and programs.

Counting the threat of viruses

- The solution for the threat of viruses is prevention.
- Detection: After infection has occurred, determine that it has occurred and locate the viruses.
- Purging: Remove the viruses from all infected systems so that the disease cannot spread further.
- * Recovery: Recover any lost data or programs.

Security Teams

- The issues of network and internet security have become increasingly more important as more businesses and people go on-line.
- Teams of people have been formed to assure in slowing hackers attacks and to disseminate.

Information on security.

→ Two secure teams are,

Computer Emergency Response Team (CERT)

Forum of Incident response and Security Team (FIRST)

Computer Emergency Response Team

+ Computer Emergency Response Team (CERT) exists as a point of contact for suspected security problem related to the Internet.

+ CERT can help determine the scope of the threat and recommend an appropriate response.

Forum of Incident Response and Security Team (FIRST)

+ Security threats are a problem that affects computers and networks around the world.

+ FIRST is made up of variety of computer emergency response teams including teams for the governments, business and academic sectors.

FIRST is made up of following Teams

+ AUSCERT

+ DEN - CERT

+ CERT - Coordinator center

+ CERT - N.L

+ CIAC

+ ANS

+ NASIRC

+ NAVLIRT

+ PCERT

+ Sunset

+ SWITCH - CERT

+ VA

2) Explain about the various approaches to intrusion

Detection and prevention

* A layered approach combining signature-based anomaly-based protocol analysis methods along with network, & host based intrusion detection and prevention systems is crucial.

Key approaches

a) Intrusion detection, prevention system (IDPS)
+ Network Intrusion detection system (NIDS).
Monitor network traffic for ~~the~~ suspicious activity. on the specific port.

b) Detection Methods

+ Signature-based detection. compares network traffic against a database of known attack signature.
+ Anomaly based detection. Identifies deviations from normal network behaviour or user activity.
+ Stateful protocol analysis. Analyze network traffic based on the state of the protocol.

c) Prevention measures

+ Firewalls: Acts as a first line of defense controlling network traffic based on predefined rules.
+ Intrusion prevention system: Can actively block or mitigate detection threats.

- * Security Information: It collects and analyzes security data from various tools to identify and respond to threats.
- * Vulnerability management: Regularly assesses and addresses vulnerabilities in systems and applications.
- * User awareness training: Educates employees about security threats and best practices.